

Software Requirements Specification for Recipe Management System

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1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for the Recipe Management System (DishDash). This specification is intended for developers, instructors, stakeholders, and evaluators involved in the development and assessment of the system. DishDash aims to automate recipe costing, unit conversions, and quotation generation for gastronomy students and culinary professionals.

1.2 Scope

DishDash is a web-based application designed to facilitate recipe creation and management, intelligent unit conversions, automatic cost calculations, and professional quotation generation for clients. The system also supports ingredient catalog management, configurable profit margins, client quotation requests, role-based access control, and calendar integration for event scheduling.

1.3 Definitions and Acronyms

- **RMS:** Recipe Management System ● **DishDash:** Name of the RMS.
- **CRUD:** Create, Read, Update, Delete
- **UI/UX:** User Interface/User Experience
- **PDF:** Portable Document Format
- **OAuth:** Open Authorization protocol
- **IVA:** Value Added Tax (Impuesto al Valor Agregado)

1.4 References

- IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- Google OAuth 2.0 Documentation

2. Overall Description

2.1 Product Perspective

- DishDash is a standalone, web-based system developed to reduce manual calculation errors and improve efficiency in recipe costing, ingredient management, and client quotation processes. The system integrates authentication services, a centralized database, and business logic components to support accurate calculations and streamlined workflows.

2.2 Product Functions

The main functions of the Recipe Management System include:

- Recipe creation, editing, and management with photo uploads
 - Intelligent unit conversion and dynamic recipe scaling
 - Centralized ingredient catalog with pricing and supplier management
 - Automated cost calculation with configurable profit margins
 - Client quotation generation and management
 - Interactive calendar for event scheduling
 - Role-based authentication (Chef / Client)
- **2.3 User Classes and Characteristics**
 - **Chef / Gastronomy Student**
 - Primary users of the system
 - Responsible for managing recipes, ingredients, costs, and quotations
 - Technical proficiency: Basic to intermediate
 - **Client**
 - Secondary users
 - Submit quotation requests and view quotation history
 - Technical proficiency: Basic
 - **2.4 Operating Environment**

- **Platform:** Web-based application
- **Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, Safari
- **Technologies:** HTML5, CSS3, JavaScript, Node.js, Express, MongoDB, third-party APIs
- **Authentication:** Google OAuth 2.0
- **2.5 Constraints**
 - An active internet connection is required for authentication and system usage
 - The system must support PDF generation for quotations
 - The database must handle concurrent user access
 - Ingredient density variations must be accurately handled during unit conversions
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3. Functional Requirements

FR-1. The system must allow users to authenticate using Google OAuth and be assigned a role (Chef or Client) with appropriate dashboard access.

FR-2. The system must maintain a centralized ingredient catalog with properties (density, equivalences, conversions), unit prices, and supplier information that can be manually updated.

FR-3. The system must allow chef to create, view, edit, and delete recipes with name, description, yield, ingredient lists (with multiple unit options), step-by-step instructions, photos, and category classification (starter, main course, dessert, beverage).

FR-4. The system must provide search and filter functionality for recipes by ingredient or category.

FR-5. The system must automatically convert between metric and imperial units, considering ingredient-specific density, and provide a conversion table for common ingredients.

FR-6. The system must scale recipe quantities proportionally when the number of portions changes, with dynamic yield recalculation.

FR-7. The system must calculate total recipe cost and cost per portion based on ingredient quantities and prices.

FR-8. The system must allow users to set configurable profit margins (per recipe or globally) and automatically suggest selling prices (cost + margin).

FR-9. The system must allow cost comparison when substituting ingredients or suppliers.

F-10. The system must allow chefs to create personalized client quotations by selecting multiple recipes, calculating total cost, taxes (IVA), and discounts automatically.

FR-11. The system must generate professional PDF quotations for download and allow quotation status tracking (pending, approved, completed).

FR-12. The system must maintain a searchable quotation history and provide quotation and client management panels in the Chef Dashboard.

FR-13. The system must allow chefs to view incoming quotation requests and respond with accept/reject decisions, proposed pricing, menu modifications, and availability confirmation.

FR-14. The system must provide a quotation request form for clients to submit detailed requests including event type, guest count, date/time, location, contact information, dietary restrictions, cuisine type, budget range, and additional notes.

FR-15. The system must allow clients to view their complete quotation request history with status updates.

FR-16. The system must include an interactive calendar interface where users can view and manage events, orders, and quotations by date.

FR-17. The system must support custom event creation (meetings, deliveries, restocking) and link quotations/recipes to specific calendar dates.

FR-18. The system must send notifications for upcoming events and deadlines, with visual distinctions for event types using colors or icons.

4. Non-Functional Requirements

NFR-1 Performance

- **NFR-1.1:** The system shall load pages within 5 seconds under normal network conditions.
- **NFR-1.2:** The system shall support at least 50 concurrent users.
- **NFR-1.3:** Unit conversions and cost calculations shall execute in less than 1 second.

NFR-2 Usability

- **NFR-2.1:** The interface shall be intuitive for users with basic technical skills.
- **NFR-2.2:** The system shall provide clear error messages and input validation.
- **NFR-2.3:** The system shall be accessible on desktop and tablet devices.

NFR-3 Reliability

- **NFR-3.1:** The system shall maintain 99% uptime during business hours.
- **NFR-3.2:** The system shall perform automatic daily data backups.

- **NFR-3.3:** The system shall handle calculation errors without data loss.

NFR-4 Security

- **NFR-4.1:** All authentication shall be handled using Google OAuth 2.0.
- **NFR-4.2:** User sessions shall expire after 24 hours of inactivity.
- **NFR-4.3:** Role-based access control shall restrict functionality by user role.
- **NFR-4.4:** Sensitive data shall be encrypted in transit and at rest.

NFR-5 Maintainability

- **NFR-5.1:** The system shall follow a modular architecture with separation of concerns.
- **NFR-5.2:** Standardized coding conventions shall be used throughout the project.
- **NFR-5.3:** The database schema shall support easy extension for new ingredient properties.